

Composable Adjudication

One interface for all on-chain adjudication.

Every protocol that needs decisions resolved integrates each adjudication system bespoke. There is no way to combine systems, and no way to start cheap and escalate only when contested.

This ERC defines a minimal standard interface, the **Adjudicator**, behind which any resolution system can sit: GenLayer validators, Kleros jurors, UMA's optimistic oracle, or a single account. Consumers (**Adjudicables**) integrate once. An adjudication is **registered** with a pointer to its definition, typically at agreement time, before any contest, like an arbitration clause. It is **activated**, with the contest evidence attached, only if a decision is actually needed. Existing systems join via thin adapter contracts; no changes required on their side.

01	Register, at agreement time	registerAdjudication(definition, data)
02	Activate, at contest time	activate(id, evidence, data)
03	Adjudicate, by any system	GenLayer, Kleros, UMA, accounts
04	Resolve, read by anyone	resolution(id) → EAS attestation

ADJUDICATOR

The lifecycle behind one interface: registered like an arbitration clause, activated only on contest, resolved as an attestation.

ESCALATION, MIX AND MATCH

§ 01

Adjudication systems differ in cost and in trust assumptions, and which is "more secure" is case-dependent, so the standard doesn't rank systems or fix an order. It provides the building blocks; whoever composes an escalation chain chooses which systems sit on it and in what sequence, escalating on contest, up to off-chain ADR.

Later rungs discipline earlier ones by their mere availability. Each stage is funded only when it is used: activation pays for the first round, and how later rounds are funded is each composite's design.

WHAT THE STANDARD COVERS

§ 03

A three-state lifecycle (Registered → Active → Resolved), payable entry points so fee-charging systems work, pull-based resolution reading, and ERC-165 discovery. Definitions, evidence, and resolutions live behind EAS attestations, so semantics evolve without touching deployed contracts. Fee semantics, evidence windows, and escalation entry points stay implementation-defined. A companion ERC covers composition and factories: pre-built, audited glue instantiated in one transaction.

COMPOSABILITY

§ 02

An aggregate of adjudicators **is** an adjudicator. An escalation router implements the same interface as its children. So does an X-of-Y panel. Because compositions are themselves adjudicators, they nest arbitrarily, and consumers never need to know whether they are talking to a single court or a whole tree of them.

- i. **Single system.** Binary resolution on one Adjudicator (GenLayer only; Kleros only).
- ii. **Escalation chains.** GenLayer to Kleros, or any sequence the composer picks, escalating on contest.
- iii. **X-of-Y panels.** 2-of-3 across GenLayer, Kleros, and UMA, or a majority of 11 agents.
- iv. **No contest, no adjudication.** Uncontested cases resolve with the default outcome; only real contests pay for adjudication.

Read the draft · github.com/genlayer-foundation/erc-composable-adjudication

Open standard · Natural-language pre-draft · Co-authored across adjudication systems

● STATUS · PRE-DRAFT V0.1 · JULY 2026